

Access signalling — Q.2931 and related standards

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This paper provides a brief description of the International Telecommunications Union (ITU-T) specified access signalling protocol Q.2931 and the supporting layer 2 signalling ATM adaptation layer (SAAL) that provides the release 1 signalling mechanism Digital Signalling System No 2 (DSS2). In addition to identifying the functions that are required to be incorporated in a signalling protocol to support call/connection establishment, in-call control and clear-down for a basic point-to-point call, the extensions required for a point-to-multipoint call are also considered. The paper then considers further extensions that have been produced to enable the negotiation, modification and look-ahead in relation to the connection in order to provide the capability set 2 step 1 DSS2 signalling mechanism. In conclusion, a simple road map to the DSS2 ITU-T standards is provided.

1. Introduction

The access signalling mechanism that has been adopted world-wide for the support of broadband ISDN (B-ISDN) in an asynchronous transfer mode (ATM) cell-based environment was the subject of a previous paper [1]. Since that paper was published, while the original mechanism produced — Digital Signalling System No 2 (DSS2) [2] — still forms the basis, signalling studies have progressed considerably. In addition, more standards bodies [3] are now studying access signalling, and a number of implementations of the initial signalling mechanism are now being offered by equipment manufacturers and are being introduced into telecommunications operators' networks. Therefore it was considered appropriate to produce an update on the current trends in B-ISDN access signalling with a brief reprise of the original, release 1, mechanism.

2. Release 1 access signalling

It was widely felt that the introduction of B-ISDN on ATM called for a far-reaching approach to be taken when studying signalling. A mechanism based on the open systems interconnection (OSI) application layer service (ALS) [4] was seen as being well suited to a B-ISDN network and the services it would support. However, there was considerable commercial pressure in some parts of the world to provide a basic broadband service by 1993/1994. Hence it was believed that any attempt to produce an advanced approach to signalling would not meet these service requirement dates. Instead it was proposed to take a two-stage approach in the specification of B-ISDN signalling:

- further studies of a signalling technique for advanced B-ISDN services to follow release 1 [5], leading to the target solution for B-ISDN signalling.

For release 1 it was agreed that the existing narrowband ISDN (N-ISDN) signalling systems should be used as a basis for the support of initial broadband services, which had relatively simple signalling requirements. Therefore for access signalling the release 1 B-ISDN signalling was based on Digital Subscriber Signalling System No 1 (DSS1) [6]. This resulted in a 'monolithic' protocol (i.e. call and bearer control bound together), but since the main requirement of release 1 signalling was to support simple point-to-point bearer services, this was not a problem. The protocol stack for release 1 B-ISDN access signalling is depicted in Fig 1.

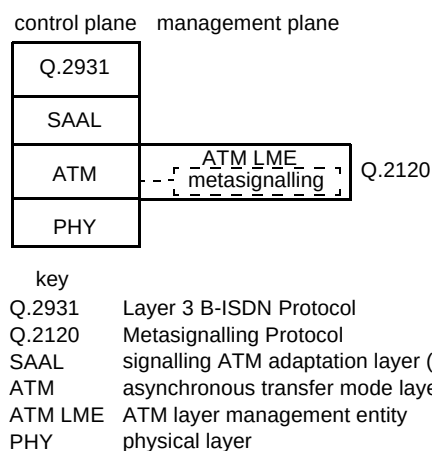


Fig 1 Protocol reference model for release 1 access signalling.

- an initial phase to meet the time-scales for the release 1 B-ISDN requirements,

2.1 Layer 2 signalling

In common with the N-ISDN and other data systems' layer 2, DSS layer 2 provides the signalling system with the signalling data link functions. These comprise the following basics:

- establishing a peer-to-peer relationship (handshake),
- transferring user data,
- error detection/control /recovery,
- link failure detection/recovery,
- congestion/flow control.

In DSS2 the signalling ATM adaptation layer (SAAL) provides the layer 2 functionality and its structure is shown in Fig 2.

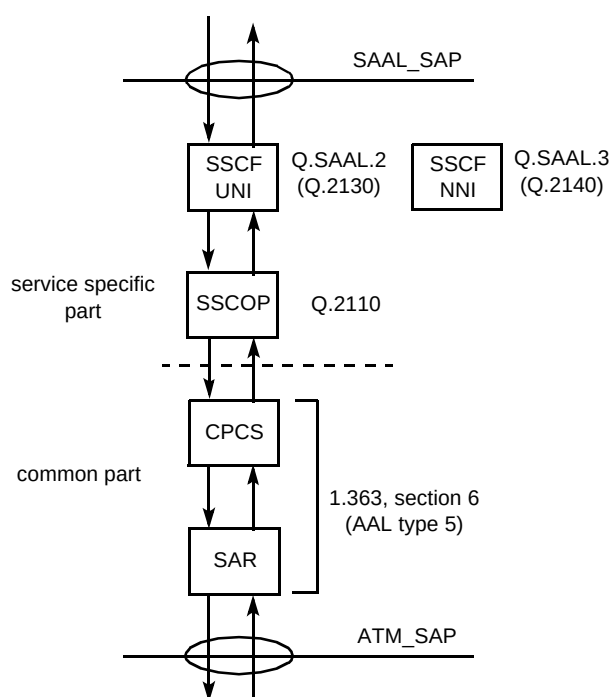


Fig 2 Structure of SAAL.

The SAAL has a common and service-specific part — the partitioning is shown by the dashed line in Fig 2.

2.1.1 SAAL — common part

The common part provides basic AAL functions using a standard AAL type which is specified in ITU-T SG 13 Recommendation I.363 [7]. For the SAAL, the common part is provided by the AAL type 5 which contains segmentation and reassembly (SAR) and the common part convergence sub-layer (CPCS). These functions provide transparent signalling data unit transport between layers and

a powerful 32-bit CRC and no further overhead is added to signalling.

2.1.2 SAAL — service specific part

The service specific part of the SAAL provides the data link layer of a signalling system comprised of service-specific connection-oriented protocol (SSCOP) [2] and signalling-specific co-ordination function (SSCF) [2].

SSCOP provides the generic layer 2 protocol functions to allow for an assured data transfer between peer entities. It is a new protocol which has been developed specifically for the SAAL.

SSCF provides the special requirements at layer 2 for matching to the user/network interface (UNI) or network/network interface (NNI) layer 3 as appropriate, which are further explained below.

SSCOP

SSCOP provides a means for the peer-to-peer transfer of information and control between the two communicating entities. The following functions are performed by SSCOP:

- sequence integrity — this preserves the order of service data units (SDUs) as transmitted by SSCOP,
- error correction — through use of the sequencing mechanism the receiving SSCOP entity can detect missing SDUs and performs error correction by selective retransmission,
- flow control — this allows the receiver to control the rate at which the peer transmitter sends information,
- error reporting to layer management — comprehensive indication of errors/problems to layer management helps to ensure the reliability of information transfer,
- keep-alive — this enables two SSCOP peer entities to remain in a link-established state even when there has been no transmission for a lengthy period,
- local data retrieval — this allows the local SSCOP user to retrieve, in sequence, data not yet transmitted or transmitted but not acknowledged,
- connection control — this allows establishment, release and resynchronisation of an SSCOP connection and transmission of variable-length unassured user-to-user messages,
- transfer of user data — this allows user data to be conveyed peer-to-peer in both assured and unassured modes,

- protocol error detection — this detects errors in protocol operation thereby providing another attribute to increase reliability,
- status reporting — this allows receiver and transmitter to exchange status information which again contributes to reliable data transfer.

SSCOP uses the following signals (primitives) to SSCF and layer management.

- SSCF interface — AA-ESTABLISH, AA-RELEASE, AA-DATA, AA-RESYNC, AA-UNITDATA, AA-RETRIEVE, AA-RETRIEVE COMPLETE,
- LME interface — MAA-ERROR, MAA-UNITDATA.

SSCOP uses two signals (primitives) to the common part — UNITDATA invoke, and UNITDATA signal.

SSCOP protocol data units (PDUs) for transmission are inserted in the UNITDATA invoke, and messages from the SSCOP peer are extracted from the UNITDATA signal.

The SSCOP protocol is supported by the following PDUs:

- establishment phase — PDUs BGN (begin) and BGAK (begin acknowledge),
- release phase — PDUs END and ENDAK (end and end acknowledge),
- resynchronisation phase — PDUs RS and RSAK (resynchronisation command and acknowledge),
- reject (part of establishment phase) — PDU BGREJ (begin reject),
- assured data transfer phase — PDUs SD, POLL, STAT and USTAT (sequenced data, status request, solicited status response and unsolicited status response),
- unacknowledged data transfer phase — PDU UD (unnumbered data),
- management data transfer phase — PDU MD (management data).

The establishment (including reject) and release phases are self-explanatory. The assured data phase sends user data in the SD PDU; if the receiver detects an out-of-sequence data unit, a USTAT PDU is returned to the transmitter to start the error-recovery process. In addition, the transmitter periodically audits the receiver by sending a POLL PDU and receiving a STAT PDU in return. Resynchronisation is used to restore consistent states at SSCOP peer entities on detection of a dislocation. Unacknowledged data transfer can be invoked by the user to transfer non-sequenced data for which there is no acknowledgement or error recovery

required. Management data phase is used to send unassured management data between peer SSCOP management entities.

SSCF UNI

Since the vast majority of the required layer 2 functions are provided by SSCOP, the SSCF at the UNI is a very 'thin' sub-layer. The service-specific co-ordination function for the UNI enables the layer 2 service expected by the UNI layer 3 to be provided by the SAAL by matching the layer 2/3 primitive interface to SSCOP. Use is made of the following primitives:

- AAL-ESTABLISH (request and indication) — carrying L3 peer-to-peer message in parameters,
- AAL-ESTABLISH (confirm),
- AAL-RELEASE (request and indication) — carrying L3 peer-to-peer message in parameters,
- AAL-RELEASE (confirm),
- AAL-DATA (request and indication) — carrying L3 peer-to-peer message in parameters,
- AAL-UNITDATA (request and indication) — carrying L3 peer-to-peer message in parameters,

ESTABLISH is used to initiate assured data transfer between AAL entities at the UNI in conjunction with point-to-point signalling virtual connections. RELEASE is used to terminate assured data transfer between AAL entities at the UNI in conjunction with point-to-point signalling virtual connections. DATA is used in conjunction with assured SDU data transfer at the UNI on point-to-point signalling virtual connections. UNITDATA is used in conjunction with unacknowledged SDU data transfer at the UNI on broadcast and point-to-point signalling virtual connections.

2.2 Metasignalling

Although, as can be seen from Fig 1, metasignalling is not part of the SAAL and therefore outside layer 2 signalling, its potential importance to B-ISDN access signalling is such that a brief explanation is required.

Unlike N-ISDN, where there is a permanent signalling 'D-channel' available, in the B-ISDN, it is possible for certain configurations of the UNI to establish a signalling channel only when required. This saving of resource could be crucial on the B-ISDN UNI where some 2000 plus terminals, requiring signalling, would seriously impinge on bandwidth available. Therefore the broadband terminal (B-TE) uses metasignalling to request the allocation of signalling resources. Metasignalling has a unique virtual channel identifier (MSVCI) and uses a series of messages (maximum length 42 octets) and procedures to assign

signalling virtual channel identifiers (SVCIs) and allocate bandwidth, monitor signalling activity and remove the SVCI and resource when no longer required. There are three phases in the procedure — assignment, checking and removal.

Metasignalling is not necessary for point-to-point UNI configurations where a fixed low-bandwidth point-to-point signalling virtual channel (PSVC), with PSVCI = 5, is used.

2.3 Layer 3 signalling

For release 1 B-ISDN services, where simultaneous control of call and bearer connection is required, the protocol has been closely based on the N-ISDN layer 3 access protocol as specified in ITU-T (CCITT) Recommendation Q.931 [6]. While there is great similarity in general, there are significant variances and a different protocol discriminator is used for the B-ISDN protocol.

Basic call and connection control for B-ISDN

As with N-ISDN access signalling there are 3 phases related to the control of the B-ISDN call and connection. These are call establishment, call clearing and in-call control. Call establishment is supported by the following messages:

- ALERTING,
- CALL PROCEEDING,
- CONNECT,
- CONNECT ACKNOWLEDGE,
- SETUP.

The necessary procedures, to enable these messages to establish a call are based on those for N-ISDN. Call clearing is supported by the following messages:

- RELEASE,
- RELEASE COMPLETE.

The procedure is unlike N-ISDN, being based on a two-message procedure and not the DISCONNECT message. 'In-call' control is supported by the following messages:

- STATUS,
- STATUS ENQUIRY.

The status procedures are similar to those produced for N-ISDN but with significant enhancement. These enhancements provide an audit function that is invoked whenever a layer 2 detects a discontinuity. In such a situation layer 2 will in addition to informing layer 2

management, also inform layer 3. Layer 3 will then invoke the status enquiry to ensure that both peer entities are aware of the current call status. There has been some slight restructuring of the messages but they basically follow that used for N-ISDN. The parameters or information elements (IEs) again draw heavily on those defined for N-ISDN. The exceptions being call reference, connection identifier, (which are in place to enable separation of call and bearer connection control to be exercised when required) and broadband bearer capability, broadband higher layer information, broadband low layer information, broadband repeat indicator and broadband sending complete, which replace the N-ISDN equivalents. In addition, there is the information required to support ATM connection characteristics, these information elements being ATM adaptation layer parameters and ATM traffic descriptors. These ATM traffic parameters will be evolved as the specification work is progressed in this area and traffic characteristics defined by SG 13 in Recommendation I.371 [8] are supported in signalling (see section 4.1). There is also a B-ISDN layer 3 compatibility mechanism based on instruction indicators which is not found in N-ISDN.

Support of N-ISDN circuit-mode services

To support 64 kbit/s-based circuit-mode services, procedures, messages and information elements are required which are additional to those for B-ISDN basic call control. There are again the three phases of signalling to control the call — establishment, clearing and in-call.

The additional messages, whose function and description are as for N-ISDN, are:

- PROGRESS,
- SETUP ACKNOWLEDGE,
- INFORMATION.

The additional information elements are the narrow-band bearer capability, narrowband low-layer compatibility and narrowband higher layer compatibility. These information elements are used with the appropriate B-ISDN messages and procedures to achieve the same results in terms of service as the N-ISDN equivalents.

Support of B-ISDN release 1 supplementary services

B-ISDN release 1 will support supplementary services which are listed in CCITT (ITU-T) Recommendation Q.767 [9] with the exception of terminal portability which is not considered to provide any useful service for the user at the B-ISDN UNI and is therefore not supported. To provide signalling support the B-ISDN access protocol includes an additional message, NOTIFY, which again is N-ISDN based and is used to convey information regarding the call/connection relating to control of the supplementary service.

3. Point-to-multipoint signalling

The support of point-to-multipoint connections had been contained in the original menu of services identified by SG 13 for release 1 signalling. However, the time constraints already mentioned led to the decision being taken to delay the completion of the study of this protocol until after point-to-point access signalling had been finalised. In the event this was not a problem since it was decided that the mechanism used for the establishment, in-call control and release of a point-to-multipoint connection [10] would be based on that used for control of point-to-point connections, release 1 signalling as in Q.2931. The possibility of using a mechanism based on ‘atomic actions’ enabling simultaneous establishment of the root connection and all leaf connections in one action was not pursued.

3.1 Scope and overview

The point-to-multipoint protocol enables a point-to-multipoint connection to be established, controlled during the in-call phase, and released from either single terminals or private networks. At present the connections established using this protocol are restricted to unidirectional transmission (i.e. root to branch).

The method of establishing the point-to-multipoint connection is for the calling (root) party to request the establishment of a connection to one of the called (leaf) parties, as in Fig 3. This makes use of the set-up procedure as for release 1 but indicates in the broadband bearer capability IE requested that the connection is a point-to-multipoint connection.

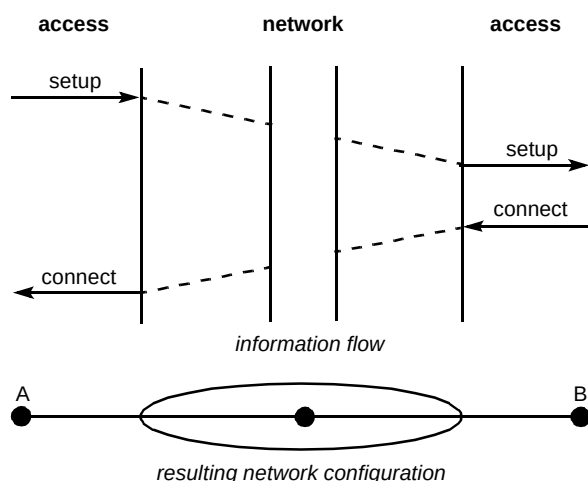


Fig 3 Establishing a point-to-multipoint call — first leaf party set-up.

Once this initial connection is active or the alerting stage has been reached, it is possible for the root to add the required leaf connections to the existing connection by sending ‘add party’ requests as in Fig 4. During the progress of the call, more leafs can be added by the use of the add party request and it is possible to use a ‘multiple add party’

request (i.e. the root can send further ‘add party’ requests without waiting for a response).

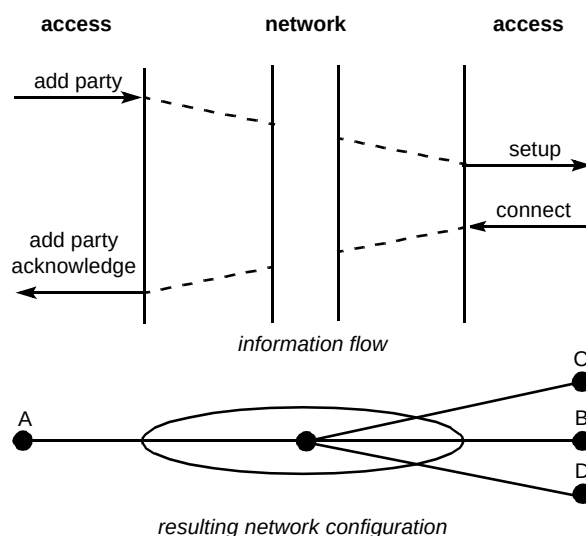


Fig 4 Establishing a point-to-multipoint call — subsequent leaf party set-up.

A leaf can be dropped at any stage during the call either as result of the root (see Fig 5) or leaf party requesting the deletion.

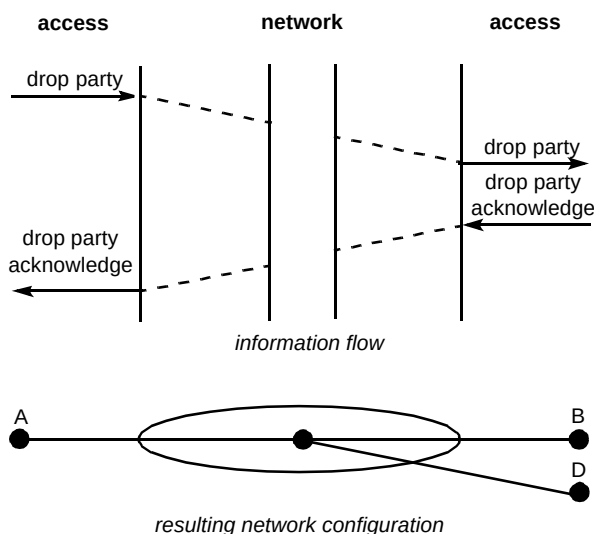


Fig 5 Dropping a party from a point-to-multipoint call.

Should a deletion result in the last leaf connection being deleted then the call is released as in Fig 6.

3.2 Messages

As the procedure makes extensive use of the point-to-point mechanism, many of the messages specified in DSS2 for release 1 in Q.2931 are reused with enhancement to cater for the point-to-multipoint environment. In addition, there are new messages and information elements specified

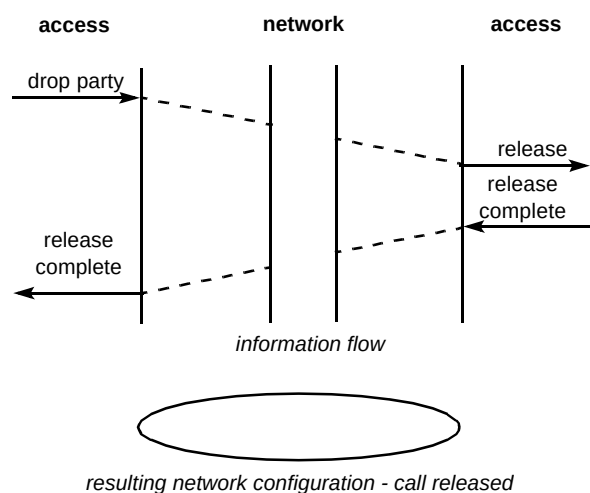


Fig 6 Dropping the last leaf connection of a point-to-multipoint call.

to support the addition and deletion of the leaf connections during the life of the call.

DSS2 (Q.2931) messages reused

The following DSS2 release 1 messages are reused with enhancement:

- ALERTING — end-point reference IE added,
- CALL PROCEEDING — end-point reference IE added,
- CONNECT — end-point reference IE added,
- SETUP — end-point reference IE added,
- STATUS — end-point reference IE and end-point state IE added,
- STATUS ENQUIRY — end-point reference IE and end-point state IE added,
- NOTIFY — end-point reference IE added.

New DSS2 (Q.2971) messages

The following new messages have been specified to support the point-to-multipoint procedure:

- ADD PARTY — sent from root to add a party to an existing connection,
- ADD PARTY ACKNOWLEDGE — sent to root from network on successful addition of party,
- PARTY ALERTING — sent to root to indicate alerting initiated,
- ADD PARTY REJECT — sent to root from network on unsuccessful request for addition of party,
- DROP PARTY — sent from root to network or network to root to drop a party from an existing point-to-multipoint connection,

- DROP PARTY ACKNOWLEDGE — sent from root to network or network to root to acknowledge a party has been dropped from an existing point-to-multipoint connection.

3.3 New information elements to support point to multipoint

Two new information elements have been specified to support point-to-multipoint signalling in addition to those in Q.2931:

- ‘end-point reference’ identifies the individual end-point in a point-to-multipoint connection and by means of a flag (set to 1) indicates if it is the first end party and therefore able to negotiate (flag set to 0 indicates subsequent party without negotiation rights),
- ‘end-point state’ indicates the state of the end-point in the point-to-multipoint connection (i.e. ADD PARTY initiated, ACTIVE, DROP PARTY initiated, etc).

3.4 Procedures supported in point-to-multipoint

The following procedures are supported in DSS2 point-to-multipoint:

- adding a party at the originating interface — set-up of first party, adding a party, action on add party received, party alerting, add party failure, add party connected, add party rejection, transit network selection,
- add party establishment at the destination interface — release 1 procedures with some modification, leaf does not support multipoint procedures,
- party dropping — exception conditions, party dropping initiated by user (root and leaf), party dropping initiated by network (root and leaf), drop collision, dropping of all parties,
- restart — enhance release 1 procedures, all parties associated with virtual channel and call shall be dropped,
- handling of error conditions — protocol discriminator error, message too short, call reference errors, end-point reference errors (missing, invalid format, procedure), message type or sequence errors, message length, general IE errors, mandatory IE errors (missing, content), non-mandatory IE errors (unrecognised), signalling AAL lost, signalling AAL released, status enquiry procedure, receipt of STATUS message,
- notification procedure — release 1 procedure enhanced for leaf and root application.

The above procedures are applicable at both the basic user interface and also for use at the interface with private

B-ISDN, with the necessary enhancements as required for interworking with private networks.

3.5 *Supplementary service interaction*

In addition to procedures to support the basic call to establish a point-to-multipoint connection, the protocol also covers the necessary interactions and subsequent modification of procedures for the release 1 supplementary services to cater for the point-to-multipoint connection environment.

4. **Capability set 2 step 1 signalling**

At the same time work was progressing on the point-to-multipoint signalling mechanism, studies were initiated to provide support for SG 13 release 2 B-ISDN requirements. Since these requirements were fairly wide ranging, it was decided to produce the protocol in a series of steps. Not only does this ease the logistics and practicalities of producing the protocol but, in adopting a stepwise or modular approach, the resulting mechanism is less complex. The aspects covered by the resulting capability set 2 step 1 (CS2-1) DSS2 signalling mechanism are:

- additional traffic parameters,
- connection characteristics negotiation during establishment,
- connection modification,
- basic look-ahead,
- multiconnection.

Although not part of the release 1 DSS2 signalling mechanism, these aspects were considered to be still pertinent to the control of the bearer connection combined with the call as well as in a separated environment. Hence CS2-1 is still considered to be part of the initial phase rather than the advanced target solution for B-ISDN signalling. Therefore the protocol for CS2-1 is again based on the DSS2 release 1. However, while this proved to be the case for the additional parameters of negotiation, modification and look-ahead, the combined call/connection approach was not appropriate for multiconnection and this is discussed further in section 5.

4.1 *Additional traffic parameters*

The DSS2 release 1 protocol provided the necessary signalling capabilities to enable the establishment, in-call control and release of a point-to-point bearer connection in an ATM network. However, the characteristics of the bearer connection that could be signalled were limited to the peak cell rate (PCR) and the required ATM adaptation layer characteristics. With regard to the former it was realised that the full flexibility of the ATM environment could not be

realised if only the PCR was signalled. Therefore the incorporation of the full range of ATM traffic parameters being defined by SG 13 [8] in DSS2 was considered a matter of priority.

Scope of additional traffic parameters in DSS2

Currently studies have progressed such that the following parameters are included in DSS2 in addition to those specified in release 1:

- statistical multiplexing, using the SCR parameter set (sustainable cell rate, maximum burst size),
- tagging option, enabling usage parameter control (UPC) function to exercise 'traffic management',
- ATC — ATM traffic characteristic,
- ABR — available bit rate,
- ABT — ATM block transfer,
- global tagging.

In the near future DSS2 will be further enhanced to contain parameters to support the following connection characteristics:

- CDVT — cell delay variation tolerance,
- quality of service.

Messages and IEs to support additional parameters

There are no additional messages to those contained in DSS2 release 1 required for the support of the additional traffic parameters. However, both the SETUP and the CONNECT have been enhanced to include the extended ATM traffic descriptor IE.

The ATM traffic descriptor IE has been extended to 30 octets in length (from the 20 octets as in release 1) in order to support the additional parameters described above.

Procedures for additional parameters

The basic procedures contained in DSS2 release 1 have been extended as follows to make use of the additional traffic parameters when establishing an ATM bearer connection:

- support of the sustainable cell rate (SCR) set — the mechanism operates on the parameter while adhering to cell loss priority, PCR limit, conditional tagging, maximum burst size (MBS), intrinsic burst tolerance, SCR and PCR relationship, SCR MBS and cell loss

priority (CLP), direction independence (all the above being dependent on the ATM layer traffic-handling capabilities),

- local support of tagging — at the originating interface the network responds to the 'tagging request' in the SETUP message, while at the destination interface the user responds to the 'tagging supported' indication in the SETUP message and if appropriate includes a 'tagging request' in the CONNECT message to be sent to the originating user,
- specific error conditions — if an ATM traffic descriptor IE is received with an invalid combination of parameters it will be treated as a mandatory IE with content error as in DSS2 release 1.

DSS2 supporting additional parameters can interwork with networks that do not support this capability but only at the release 1 level; it is not possible to interwork the additional parameters with N-ISDN.

The additional parameter capabilities have no impact on the supplementary services as specified for DSS2 release 1.

4.2 Negotiation

While the additional parameters included in DSS2 make it possible to vary the characteristic of the connection being established, the release 1 procedures still meant that the connection was fixed at establishment and for the duration of the call. Therefore a major step forward in signalling was seen as the ability to negotiate regarding the ATM connection characteristics required at the time of establishment.

Scope and overview

The negotiation capability within DSS2, when used in conjunction with the DSS2 release 1 signalling mechanism and the additional traffic parameters that have been specified, enables the characteristics of the connection to be negotiated at the time of establishment. In addition to its application to the point-to-point connection it can, when applied with the DSS2 point-to-multipoint protocol, enable the characteristics of the connection established to the first leaf party to be negotiated.

The negotiation capabilities are only applicable during establishment, catering for the following:

- negotiation of connection characteristics by providing alternative traffic descriptors,
- negotiation of cell rate traffic parameters using a minimum traffic descriptor.

With the alternative traffic descriptor IE, parameters are handled as a single entity. When the minimum traffic descriptor IE is used, a range of parameter values are allowed and handled independently. Traffic parameter negotiation is possible with the alternative ATM traffic descriptor IE, whereas it is only possible to negotiate the peak cell rate when the minimum ATM traffic descriptor IE is used (see Fig 7).

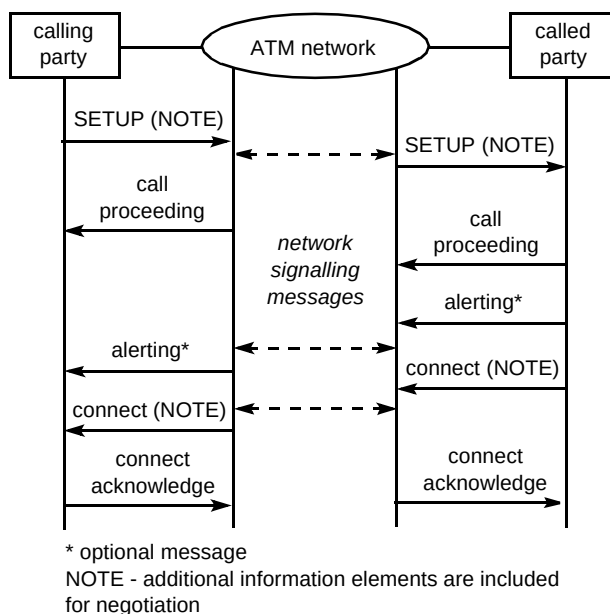


Fig 7 Negotiation of additional traffic parameters leading to successful connection establishment.

Messages and information elements

There are no new messages specified to support DSS2 negotiation; however, two DSS2 release 1 messages are used with enhancements as follows:

- SETUP — either alternative ATM traffic descriptor IE or minimum ATM traffic descriptor are included (but not both),
- CONNECT — ATM traffic descriptor IE (included to indicate which traffic parameters have been successfully negotiated).

Two new information elements have been specified in addition to those contained in DSS2 release 1 to enable negotiation to be supported:

- alternative ATM traffic descriptor — this specifies the alternative ATM traffic descriptor to be used for negotiation of traffic parameters during connection establishment (can have any combination of parameters allowed for in the ATM traffic descriptor for the specified bearer capability; alternative bandwidth must be less than originally requested),

- minimum acceptable ATM traffic descriptor — this specifies the minimum acceptable ATM traffic parameters in the negotiation of traffic parameters at set-up (these are the lowest values that the user will accept).

Procedures supported

The following procedures are supported in DSS2 negotiation:

- negotiation of connection characteristics (originating interface) — negotiation request, negotiation traffic parameter (minimum acceptable or alternative ATM traffic), negotiation acceptance,
- negotiation of connection characteristics (destination interface) — negotiation request, negotiation traffic parameter (minimum acceptable or alternative ATM traffic), negotiation confirmation.

If the negotiation is successful, a CONNECT message is sent containing the ATM traffic descriptor agreed; if unsuccessful, a RELEASE COMPLETE will be returned to originating interface.

Negotiation is uniform at both S_B/T_B (reference) and T_B (reference) interfaces and is terminated at the interworking points with other networks. There are no interactions with the release 1 supplementary services.

4.3 Modification

With the ability to negotiate additional traffic parameters at connection establishment, the CS-2 enhancement was further advanced towards the fully flexible access signalling system that was required for B-ISDN if the full capabilities afforded by the ATM environment were to be realised. However, from the outset it was realised that, unlike the N-ISDN call dependent on a 'fixed' 64-kbit/s switched environment, B-ISDN calls would be far more dynamic, the characteristics of the supporting connections being capable of being varied throughout the lifetime of the call to meet the end user's demand. Therefore the introduction of in-call modification of the connection was seen as being a crucial aspect of DSS2.

Scope and overview

The connection modification capability within DSS2, when used in conjunction with the DSS2 release 1 signalling mechanism and the additional traffic parameters that have been specified, enables the characteristics of the connection to be modified during the progress of the call to meet the end user's requirements. Currently it is only the PCR that can be modified, but the mechanism is sufficiently generic to enable all the connection attributes (see section

4.1) to be modified eventually. Future developments will see negotiation also being introduced into the modification process. The ability to modify a connection is at present restricted to the connection owner and is only possible with point-to-point connections.

In addition, it is only possible to request modification to those parameters originally requested in a given direction. However, it is possible to request an increase in one direction with a decrease in the opposite direction in one modification request. Connections must be established before they can be modified. The mechanism cannot be applied to connections that are in the process of being established or cleared. Clearing will always take priority over modification; hence, if during the modification process clearing is invoked, the connection will be cleared down without the modification being completed.

When initiating a modification request to increase or decrease the PCR, the initiating user (connection owner) must:

- be prepared to receive traffic with an ATM traffic parameter that is the greater of the unmodified or modified backward parameter for the connection being modified,
- be prepared to transmit traffic with an ATM traffic parameter that is the lesser of the unmodified or modified forward parameter for the connection being modified.

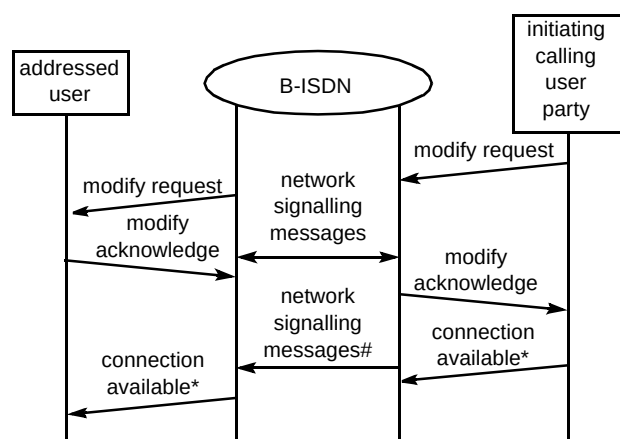
It is also possible to modify in the given direction the PCR for $CLP = 0$ or $CLP = 0 + 1$ in the same modification request. All parameters must be decreased or all must be increased.

The modification process also allows the connection to continue to support the service application.

Procedures are also contained that allow for rejection of the modification request by the 'non-connection owner' and confirmation to the 'connection owner' and 'non-connection owner' that modification has been successful (see Figs 8—10).

Messages

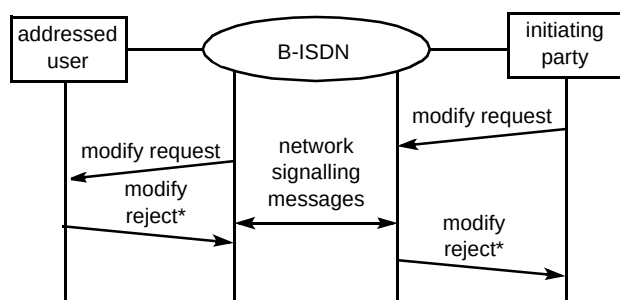
The procedure, while it is an enhancement to the DSS2 release 1 point-to-point mechanism, is in fact an embedded mechanism. Therefore there are new messages and information elements specified to support the requested modification of connections during the life of the call.



* This message is sent only if requested by the addressed user by including a broadband report type information element in the MODIFY ACKNOWLEDGE message.

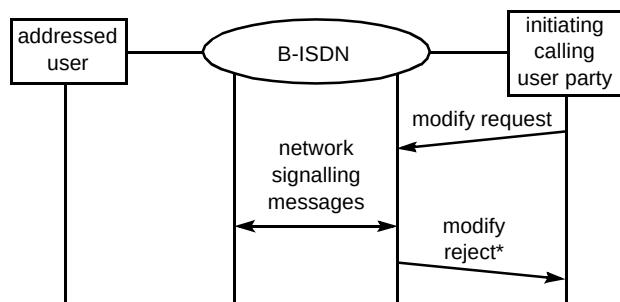
Network signalling messages for support of the information contained in the DSS2 CONNECTION AVAILABLE message. Not required if connection available information is not requested by the addressed user.

Fig 8 Successful modification.



* Shall contain cause IE indicating why modification was unsuccessful

Fig 9 Unsuccessful modification — rejection by called user.



* Shall contain cause IE indicating why modification was unsuccessful

Fig 10 Unsuccessful modification — rejection by network.

The following new messages have been specified to support the modification procedure:

- **MODIFY REQUEST** — sent from initiating user to network and network to called user to request modification of connection,

- **MODIFY ACKNOWLEDGE** — sent to initiating user from network to indicate modification request is accepted,
- **MODIFY REJECT** — sent to initiating user from network to indicate modification request is not accepted,
- **CONNECTION AVAILABLE** — sent from initiating user to network or network to 'non-connection owner' to confirm modification, if requested in MODIFY ACKNOWLEDGE.

Information elements

The modification mechanism in DSS2 reuses many of the information elements that have been specified to support point-to-point connection control signalling. In addition the following new information element has been specified for use in MODIFY ACKNOWLEDGE message:

- 'broadband report type' is included in the MODIFY ACKNOWLEDGE message when the 'non-connection owner' requires confirmation of modification success from the 'connection owner' — confirmation is sent using the CONNECTION AVAILABLE message (this is considered to be a generic IE and will probably have wider application).

Procedures supported

The following procedures are supported in DSS2 connection modification:

- procedures at requesting entity — modification request, modification acknowledgement, indication of modification rejection, response to STATUS while in modify request state, no response to modification request,
- procedures at responding entity — modification indication, modification acceptance, modification confirmation, modification rejection,
- transit entity conveyance of CONNECTION AVAILABLE — transparently while in active state.

The above procedures are applicable at both the basic user interface and also for use at the interface with private B-ISDN. There is no interworking with other networks.

There are no identified interactions with the release 1 supplementary services.

4.4 Look-ahead

The basic look-ahead capability was introduced into DSS2 CS2-1 mainly to allow the network to determine the

viability of establishing a connection. Although it is possible that in future this capability may be extended for user-to-user application, other developments (see section 5) may make this unnecessary. The look-ahead capability in DSS2 makes use of the generic functional protocol (GFP) as specified for B-ISDN [11], which is based closely on the N-ISDN DSS1 mechanism specified in Recommendation Q.932. The B-ISDN GFP using connection-oriented bearer-independent (COBI) techniques enables:

- activation and deactivation,
- invocation and operation,
- interrogation,
- status request,
- status notification,

of a process/capability/supplementary service in association with or outside an existing call. A brief outline of messages, information elements and procedures of GFP is contained in the following description of look-ahead.

Overview and scope of look-ahead

The basic DSS2 look-ahead allows the incoming side of the network (in response to a request from the calling user) the capability of looking ahead to the outgoing side of the network, for example, to ascertain the viability of supplying the network service requested by the 'calling user'. The decision as to whether the look-ahead capability should be invoked is left to the incoming gateway exchange. It is also possible for a private B-ISDN to indicate to the public B-ISDN that basic look-ahead is required.

Further examples of how look-ahead in DSS2 can be applied are:

- to check whether the terminal addressed is compatible or incompatible,
- to check whether the terminal addressed is free or busy,

and can be used prior to the establishment of the call connection (see Figs 11 and 12). However, the application of DSS2 look-ahead is not part of the basic DSS2 signalling service which only defines the generic procedure.

The procedure makes extensive use of the point-to-point mechanism; therefore many of the messages specified in DSS2 for release 1 in Q.2931 are reused with enhancement to cater for the point-to-multipoint environment. In addition, there are new messages and information elements (IEs) specified to support the addition and deletion of the leaf connections during the life of the call.

Look-ahead messages

The procedure makes use of the point-to-point mechanism and the GFP. Therefore messages specified in DSS2 for release 1 in Q.2931 and for DSS2 GFP in Q.2932 are reused with enhancement to cater for look-ahead. In addition, there is a new information element specified for look-ahead together with look-ahead operations and errors coded in ASN.1.

The following DSS2 release 1 messages and GFP message are reused with enhancement:

- **SETUP** — network look-ahead indicator IE to indicate capability used,
- **CALL PROCEEDING** — as for release 1, sent to calling party to indicate look-ahead initiated by network,
- **FACILITY** — carries look-ahead invoke to called party and look-ahead return result from called party.

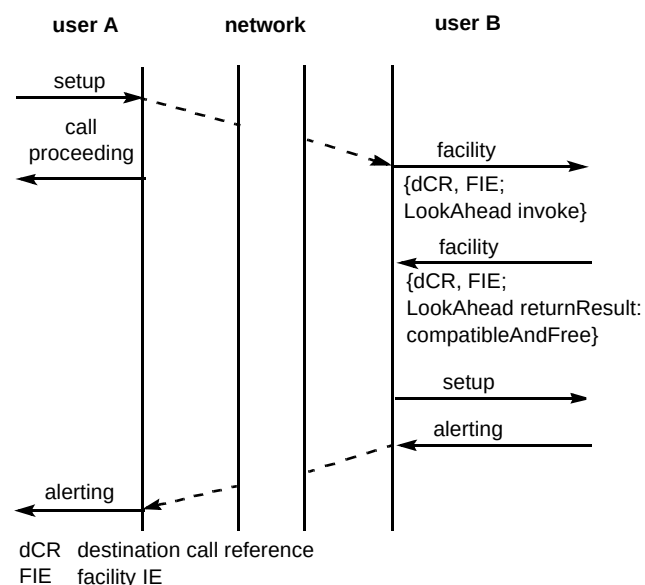


Fig 11 Look-ahead — positive answer.

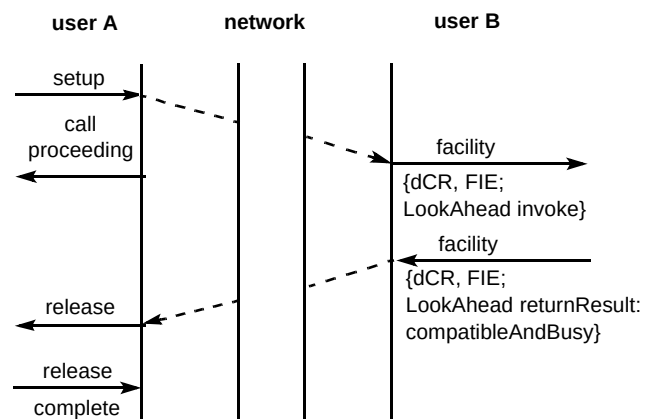


Fig 12 Look-ahead — negative answer.

Information elements

The look-ahead mechanism in DSS2 uses an information element that has been specified to support GFP signalling. In addition, the following new information element has been specified for use in the SETUP message (at T_B reference):

- network look-ahead is included in the SETUP (at T_B reference) message to indicate when a calling network has used the look-ahead capability in that call,
- facility is used in FACILITY message to convey the operations for look-ahead at the destination interface.

Procedures supported

The following procedures are supported in DSS2 look-ahead at the S_B reference:

- normal operations — send look-ahead invoke, start look-ahead timer, enter look-ahead state, stop timer/terminate procedure,
- exceptional procedures — stop timer/terminate procedure.

The following procedures are supported in DSS2 look-ahead at the T_B reference:

- originating T_B interface normal operations — send Look-AheadTb invoke, start look-ahead timer (T_b), forward result to application, stop timer T_b /terminate procedure,
- originating public network normal operations — forward invocation to user B, enter look-ahead state, send result, terminate procedure,
- exceptional procedures public network — stop timer/terminate procedure, for AAL-EST receipt abort procedure, for AAL-REL receipt ignore and remain in current state,
- destination public network normal operations — send Look-AheadTb invoke to private network, start look-ahead timer (T_b), enter look-ahead state, forward result, stop timer T_b /terminate procedure,
- destination T_B interface normal operations — forward invocation to user B, enter look-ahead invoked state, send result to public network, terminate procedure,
- exceptional procedures public network — stop timer T_b /terminate procedure, for AAL-REL receipt abort procedure, for AAL-EST receipt ignore and remain in current state.

Using the GFP precludes interworking with non-ISDNs, Frame Relay and packet switched public data networks (PSPDNs).

5. Multiconnection signalling

As with the other signalling Recommendations constituting capability set 2 step 1 in ITU-T SG 11, it was intended that the signalling protocol to support multiconnection should be based on the CS-1 protocol Recommendation Q.2931. However, initial studies identified that the monolithic structure of Q.2931 prevented the full required capability of signalling support for a point-to-point multiconnection call from being realised. Using Q.2931 it is not possible to add or delete connections to a call as required since the call and connection are bound together.

Therefore the protocol required for the support of the multiconnection call requires a separation of the call and bearer control functionality to enable the degree of flexibility in control required to be achieved. In order to achieve simplicity and efficiency in the control mechanism for multiconnections, separate protocols for call and connection control are considered the best approach. This approach also allows the possibility of prenegotiation of the connections to be added and a separate protocol is specified for this purpose (see section 5.3).

5.1 Connection control

It has been possible to adapt Q.2931 to provide a connection control protocol and this is currently under study in ITU-T SG 11. Use is made of many of the messages developed for Q.2931 and also the information elements used in the CS-1 DSS2 protocol. However, since this is a connection control mechanism rather than a call and connection control mechanism, any IE, message or procedure that is call control related has been stripped out. Therefore compatibility with Q.2931 is not possible.

5.2 Call control

Study of a mechanism for call control in the separated environment has also been undertaken in ITU-T SG 11 using Q.2931 as the basis. While this appeared to be an adequate solution for the simple point-to-point multiconnection call, it became apparent that it would not satisfy the more stringent requirements for the multipoint/multiconnection environment. Since it is obvious that such a multipoint/multiconnection capability will be required in future networks, a number of organisations including the European Computer Manufacturers Association (ECMA), the European Telecommunications Standards Institute (ETSI) and BT decided that, rather than use a short-term expedient solution (Q.2931 based), a full solution is required. Therefore studies have started on a call control

mechanism based on the remote operations service element (ROSE-) based generic functional protocol as specified in Recommendation Q.2932. More detailed information on this control protocol can be found elsewhere [5].

5.3 Prenegotiation

In addition to call and connection control it was realised that, with separation of call and bearer connection, it was now possible to negotiate ahead of establishment the characteristics of a connection that was to be associated with a call (except in the case of simultaneous call and connection establishment). This mechanism, known as pre-negotiation, allows end users to check compatibility with the network, as well as compatibility and availability at the remote user with regard to the one or more connections they intend to establish within the lifetime of the call. The protocol is based on the generic functional protocol and the operations involved fulfil the requirements of pre-negotiation without the reservation of any connection-oriented resources within the network.

6. Conclusions

This paper has identified how access signalling protocols have been developed from their N-ISDN predecessor to enable the support of the initial capabilities required for the support of point-to-point connections. Their enhancement to enable the further network requirements for the support of the capability set 2 functions has also been covered.

These initial phases of the B-ISDN access signalling protocol DSS2 have formed a valuable basis, not only allowing operators and manufacturers to provide initial broadband control, but also as a starting point for more advanced studies. However, with the need to move into the multipoint/multiconnection network control environment, the scope of the initial solution to B-ISDN access signalling has reached its limit. Therefore the studies into the separated call and bearer control environment are now vital if a viable solution to the control of advanced networks and the required access signalling support is to be realised.

Appendix

ITU-T access signalling standards

A1 B-ISDN signalling standards

The standards in this section relate to the access and inter-exchange signalling and cover the data link layer and the network (call control) layer for both basic circuit-switched calls and supplementary services. The standards defining the access signalling protocols cover both the network and terminal (S_B/T_B or T_B interface) aspects within the same standard. The status of the standards is given in Tables A1-A7 as at January 1998.

A2 Layer 2 user/network interface protocol

The layer 2 protocol, signalling ATM adaptation layer (SAAL), standard is contained in ITU-T Recommendations Q.2100, Q.2110 and Q.2130 which have been approved by SG 11 (see Table A1).

In order to complete the B-ISDN (release 1) access signalling protocol standards, a suite of standards covering the conformance test specification for layer 2 of access signalling is being drafted. This work has not yet started in ITU-T. These standards specify the protocol implementation conformance statement (PICS) proforma, protocol implementation extra information for testing (PIXIT), and the abstract test suite (ATS) specified using the TTCN language.

In Tables A1-A7, which give the details of the standards and their status, the following key is used for the status column:

- 0 — Status of draft is unknown
- 1 — No draft available
- A/R — awaiting review
- C — cooling
- F — frozen
- V — voting (post-initiation of resolution 1 (determination))
- A — approved (by SG 11)

Table A1 Layer 2 user/network interface standards.

Title	Number	Status	Comments
B-ISDN signalling ATM adaptation layer (SAAL) overview	Rec Q.2100	A	Technically agreed, approval by SG 11 3/94#
B-ISDN signalling ATM adaptation layer — service specific connection-oriented protocol (SSCOP)	Rec Q.2110	A	Technically agreed, approval by SG 11 3/94#
B-ISDN signalling ATM adaptation layer — service-specific co-ordination function for support of signalling at the user-to-network interface (SSCF at UNI)	Rec Q.2130	A	Technically agreed, approval by SG 11 3/94#
AAL 5	Rec I.363	A	Approved at January 93 meeting of SG 13

Table A2 Details for release 1 — layer 3 user/network interface standards basic call.

Title	Number	Status	Comments
B-ISDN user/network interface layer 3 protocol addendum for ATM traffic characteristics	Rec Q.2931 add 1	A	Technically agreed and frozen by SG 11/1. Resolution 1 approval by SG 11 01/97, refers to traffic characteristics as specified in Q.2961 and aligned with Rec I.371.
B-ISDN user/network interface layer 3 protocol — living list	Rec Q.2931 Ed 2	C	Baseline text reaching a degree of stability. As over 2 years since approval of Iss 1, Ed 2 may be sent for determination in 1998.
B-ISDN user/network interface layer 3 protocol	Rec Q.2931	A	Technically agreed approved by SG 11 9/94

Table A3 Details for capability set 2 step 1 (CS2-1) — layer 3 user/network interface standards extension to basic call to support CS 2-1.

Title	Number	Status	Comments
Additional traffic parameter indications	Rec Q.2961.1	A	Although technically frozen at 9/94 required considerable reworking to align with stable text in Rec I.371. Approved at SG 11 05/95. Parts 2—6 See Table A4.
Negotiation of connection characteristics (traffic parameters) during call establishment	Rec Q.2962	A	Document approved at the 02/96 meeting of SG 11.
Modification of traffic parameters during the active phase	Rec Q.2963.1	A	Progress is being made; part of document (PCR) approved by SG 11 at the 02/96 meeting. Part 2 (SBR & MBS) See Table A4.
Look-ahead capability	Rec Q.2964	A	Document approved by SG 11 at the 02/96 meeting. Position on whether any further parts to this document will be produced is still under study.
Generic protocol mechanisms to support supplementary services	Rec Q.2932.1	A	Progress made; part of document was SG 11 approved at 02/96. Remaining parts dealing with COBI (see Table A4).
Point-to-multipoint protocol	Rec Q.2971	A	Approved at SG 11 meeting 05/95.
Point-to-multipoint protocol — living list	Rec Q.2971 Ed 2	C	Baseline text reaching a degree of stability. Since 2 years approval of Iss 1, Ed 2 may be sent for determination in 1998.
Support of point-to-point multiconnection calls	Rec Q.298x Rec Q.298bc Rec Q.298cc (ROSE) Rec Q.298xx (2931 ext) Rec Q.298pn	C C C C	Document structure considerably modified. Document now to be produced as separate parts — bearer connection control, call control and pre-negotiation. All parts of Q.298x are now in the cooling state. It has been decided that there will be parallel studies of 298cc (ROSE) and Q.298xx (2931 ext).

A3 Layer 3 user/network interface standards — basic call

Layer 3 protocol for B-ISDN release 1 call control is contained in ITU-T Recommendation Q.2931 which has recently been agreed by SG 11/2 and submitted to SG 11 for approval by the resolution 1 procedure (see Tables A2, A3, A4 and A5).

In order to complete the B-ISDN (release 1) access signalling protocol standards, a suite of standards covering the conformance test specification for the basic call is being drafted. This work has not yet started in ITU-T.

These standards specify the protocol implementation conformance statement (PICS) proforma, protocol implementation extra information for testing (PIXIT) and the abstract test suite (ATS) specified using the TTCN language.

A4 Common aspects of B-ISDN application protocols for access signalling and network signalling interworking

The standards for common aspects of B-ISDN release protocols, i.e. cause values, interworking between DDS 2 and B-ISUP, are contained in ITU-T Recommendations Q.2610 and Q.2650 which have been agreed by SG 11/2 and approved by the resolution 1 procedure by SG 11 in September 1994 (see Table A6).

A5 B-ISDN (release 1) supplementary services

The supplementary services to be supported in B-ISDN release 1 are as listed in Recommendation Q.767 (N-ISDN UP for the international interface) [9]. The services identified are:

Table A4 Details for capability set 2 step 2 (CS2-2) — layer 3 user/network interface standards extension to basic call to support CS2-2.

Title	Number	Status	Comments
Support of frame relay services in B-ISDN	Rec Q.2933	A	Approved by SG 11 at 02/96.
Call priority	Rec Q.2959	A	Document approved by SG 11 at 02/96.
Additional traffic parameter indications ATM transfer capabilities (ATC)	Rec Q.2961.2	A	Approved by SG 11 at 01/97 meeting
Additional traffic parameter indications available bit rate (ABR)	Rec Q.2961.3	A	Approved by SG 11 at 09/97 meeting.
Additional traffic parameter indications bulk transfer (ABT)	Rec Q.2961.4	A	Approved by SG 11 at 09/97 meeting
Additional traffic parameter indications cell delay variation tolerance (CDVT)	Rec Q.2961.5	F	Minor technical problems still needing consultation with SG 13; expected to be submitted for determination by SG 11/1 at 05/98 meeting
Additional traffic parameter indications global tagging	Rec Q.2961.6	V	Submitted for determination by SG 11/1 at 01/98 meeting
Additional traffic parameter indications quality of service (QoS)	Rec Q.2961.7	C	Technical problems still needing consultation with SG 13; possibly to be submitted for determination by SG 11/1 at 01/98 meeting
Negotiation of connection characteristics (PCR, MBS and SCR) during call establishment	Rec Q.2962 Ed 2	V	Submitted for determination by SG 11/1 at the 01/98 meeting
Modification of traffic parameters (PCR, MBS and SCR) during the active phase	Rec Q.2963.2	A	Approved by SG 11 at the 09/97 meeting.
Coding examples	Rec Q.2939.1	A	Approved by SG 11 at the 09/97 meeting. This recommendation is closely based on the N-ISDN counterpart Q.939.
Generic protocol mechanisms to support supplementary services (connection-oriented bearer-independent (COBI))	Rec Q.2932.2	C	Minor technical problems but possibly to be submitted for determination by SG 11/1 at 01/98 meeting
Connection modification with negotiation of traffic parameters during the active phase	Rec Q.2963.3	V	Submitted for determination by SG 11/1 at the 01/98 meeting
Support of supplementary services on B-ISDN (ECT)	Rec Q.295x	1	This activity has been given a fairly low priority by SG 11. Since the sponsoring organisations are not currently attending SG 11, this item has been shelved.
B-ISDN user/network interface layer 3 protocol for switched virtual path	Rec Q.2934	V	Studies have been progressed and submitted for determination at 01/98 meeting.

Table A5 Details for capability set 2 step 3 (CS2-3) — layer 3 user/network interface standards extension to basic call to support CS 2-3.

Title	Number	Status	Comments
Generic concepts for the support of multipoint and multiconnection calls	Rec Q.293x	C	Good progress has been made and it is expected that this document may be a candidate for determination at the 05/98 SG 11 meeting.
DSS 2 generic ID transport	Rec Q.2941.1	A	Approved by SG 11 at the 09/97 meeting.
Leaf initiated join	Rec Q.29aa	A/R	Editor's draft text available for comment and study at the 09/97 SG 11 meeting.
3rd party control	Rec Q.29bb	1	Work initiated at 10/96 meeting but no text yet available.
Proxy signalling	Rec Q.29cc	1	Work initiated at 10/96 meeting but no text yet available
Lightweight protocol	Rec Q.29dd	1	Work initiated at 10/96 meeting but no text yet available. BT are pressing for the DAVIC DSM-CC to be used as a basis (BT has been requested to act as editor).
See CS 2-2 (Table A4)			

- direct dialling in (DDI),
- multiple subscriber number (MSN),
- calling line identification presentation (CLIP),
- calling line identification restriction (CLIR),
- connected line identification presentation (COLP),
- connected line identification restriction (COLR),
- sub-addressing (SUB),
- terminal portability (TP),
- closed user group (CUG),
- user-to-user signalling implicit (UUSI).

Table A6 Common aspects of B-ISDN access and network signalling and interworking.

Title	Number	Status	Comments
Use of cause and location in B-ISDN and B-ISUP	Rec Q.2610	A	Technically agreed approval by SG 11 9/94
Interworking between B-ISUP and B-ISDN access signalling (DSS2, L3)	Rec Q.2650	A	Technically agreed approval by SG 11 9/94

Table A7 Details of B-ISDN (release 1) supplementary services standards.

Title	Number	Status	Comments
Release 1 supplementary services	Rec Q.2951 Ed 2	F	Baseline text reaching a degree of stability. As over 2 years since approval of Iss 1 (not CUG), Ed 2 may be sent for determination in 1998.
Direct-dialling-in (DDI) (DSS2)	Rec Q.2951.1	A	Approval by SG 11 9/94
Direct-dialling-in (DDI) (B-ISUP)	Rec Q.2761-4	A	In B-ISUP DDI is supported in the basic call which is technically approved by SG 11 9/94
Multiple subscriber number (MSN) (DSS2)	Rec Q.2951.2	A	Approval by SG 11 9/94
Multiple subscriber number (MSN) (B-ISUP)	Rec Q.2761-4	A	In B-ISUP MSN is supported in the basic call which is technically approved by SG 11 9/94
Calling line identification presentation (CLIP) (DSS2)	Rec Q.2951.3	A	Approval by SG 11 9/94
CLIP (BISUP)	Rec Q.2731.3	A	Approval by SG 11 9/94
Calling line identification restriction (CLIR) (DSS2)	Rec Q.2951.4	A	Approval by SG 11 9/94
CLIR (BISUP)	Rec Q.2731.4	A	Approval by SG 11 9/94
Connected line identification presentation (COLP) (DSS2)	Rec Q.2951.5	A	Approval by SG 11 9/94
COLP (BISUP)	Rec Q.2731.5	A	Approval by SG 11 9/94
Connected line identification restriction (COLR) (DSS2)	Rec Q.2951.6	A	Approval by SG 11 9/94
COLR (B-ISUP)	Rec Q.2731.6	A	Approval by SG 11 9/94
Sub-addressing (SUB) (DSS2)	Rec Q.2951.8	A	Approval by SG 11 9/94
SUB (B-ISUP)	Rec Q.2731.8	A	Approval by SG 11 9/94
Terminal portability (TP) (DSS2)	Rec Q.2953.4	F	Technically agreed and frozen by SG 11/2. Resolution 1 approval initiated, approval deferred by SG 11 since TP is not considered to be applicable to B-ISDN release 1.
TP (B-ISUP)	Rec Q.2733.4	F	Technically agreed and frozen by SG 11/2. Resolution 1 approval initiated, approval deferred by SG 11 since TP is not considered to be applicable to B-ISDN release 1. Interworking with N-ISDN TP is supported.
Closed user group (CUG) (DSS2)	Rec Q.2955.1	A	Approval by SG 11 01/97
CUG (B-ISUP)	Rec Q.2735.1	A	Approval by SG 11 01/97
User-to-user signalling implicit (UUSI) (DSS2)	Rec Q.2957.1	A	Approval by SG 11 9/94
UUS 1 (B-ISUP)	Rec Q.2737.1	A	Approval by SG 11 9/94

The stage 1 and 2 definitions as specified for N-ISDN were considered appropriate for use in the stage 3 definition for release 1 B-ISDN and were used (where available).

Although CUG is contained in the SG 13 release 1 timetable, it has not been possible to complete the standards to meet the release 1 signalling standards approval deadline.

Therefore the standard for this supplementary service has been deferred to CS-2 (B-ISDN).

In order to complete the B-ISDN (release 1 access signalling protocol standards), a suite of standards covering the conformance test specification for each supplementary service is being drafted (see Table A7). This work has not yet started in ITU-T. These standards specify the protocol

implementation conformance statement proforma, protocol implementation extra information for testing, and the abstract test suite which are specified using the TTCN language.

References

- 1 Law B: 'Signalling in the ATM network' BT Technol J, 12, No 3, pp 93—107 (1994).
- 2 ITU-T Recommendations: 'Q.2110, Q.2130 and Q.2931 (DSS2 Release 1 [Access] Signalling)', (1994).
- 3 Law B: 'B-ISDN signalling standards', BT Technol J, 16, No 2, pp 11—19 (April 1998).
- 4 ITU-T Recommendation Q.1400: 'Architecture framework for the development of signalling and OAM protocols using OSI concepts', (1993).
- 5 Knight R R and Law B: 'The call control protocol in a separated call and bearer environment', BT Technol J, 16, No 2, pp 75—86 (April 1998).
- 6 ITU-T Recommendations: 'Q.920, Q.921, Q.930 and Q.931', (1993).
- 7 ITU-T Recommendation I.363: 'ATM adaptation layer', (SG13) (1993).
- 8 ITU-T Recommendation I.371: 'Traffic control and congestion control in the B-ISDN', (SG13) (1996).
- 9 ITU-T Recommendation Q.767: 'Application of the ISDN user part of CCITT SS No 7 at the international interface', (SG11) (1991).
- 10 ITU-T Recommendation Q.2971: 'DSS2 layer 3 specification for point-to-multipoint call/connection control', (SG11) (1995).
- 11 ITU-T Recommendation Q.2932: 'Generic functional protocol', (SG11) (1996).



Bryan Law joined BT as an apprentice in 1958. For the next five years he was employed in subscriber apparatus maintenance and installation and then strowger exchange maintenance duties. He was then involved in transmission repeater station construction and maintenance duties at the BT Tower Television Switching Centre. In 1967 he took on the responsibility for trunk cable installation standards and then commissioning of FDM and digital transmission equipment.

Having been involved in the world of transmission for seventeen years he returned to his former field of switching in 1980 when joining the Telecommunications Strategic Studies department undertaking information flow studies for System X and then being responsible for the specification of CCITT SS No 7 for use in the BT network. During this period he represented BT at the international standards bodies meetings responsible for the study of CCITT SS No 7 and became the UK co-ordinator for the studies in this area. He joined his present group in 1989 to work on the RACE project studying access signalling for the support of B-ISDN. He currently represents BT in ETSI and ITU-T studies of B-ISDN signalling and is responsible for co-ordinating BT activity in the area of access signalling. He is the UK co-ordinator for B-ISDN signalling studies.



Dick Knight joined BT as an apprentice in 1971 and moved to BT Laboratories in 1979. He worked on System X development in the areas of the test network subsystem and customer line card design acceptance testing. In 1986 he was made responsible for the software to implement a programmable ISDN signalling design acceptance tester. He designed and implemented a protocol to collect network management information for the Jetphone network. From 1993 to 1994 he moved onto broadband signalling design by managing the collaborative RACE MAGIC project on behalf of twelve European companies — a project whose results are still having one of the largest impacts on broadband signalling design in the world. With a short break in 1996 to design and patent an application protocol to collect information such as burglar alarms over the ISDN D-channel packet access, he has continued to work on various aspects of broadband signalling, both internal projects and representing BT at ETSI with the occasional representation at ITU-T to contribute some of the MAGIC results.